

HEALTHCONNECT CLINIC

Week 6 — Advanced Analytics & Decision Support Report

Week 6 deepens and validates the strongest Week 5 findings rather than repeating the full exploratory analysis. The analysis focuses on previous no-show history, reminder patterns, appointment type, KPI validation, and decision support.

1. EXECUTIVE SUMMARY

The Week 6 Power BI analysis confirms that missed appointments remain a major operational issue. The baseline dashboard reports a 48.46% no-show rate and a 46.28% attendance rate, with 5.26% cancelled appointments. The most important segment-level signal is previous no-show history: the no-show rate rises across the history groups and reaches approximately 97% for patients with five previous no-shows in the displayed analysis.

Appointment type also shows a meaningful segmentation pattern. Follow-up appointments have the highest no-show rate at 51.23%, followed by Diagnostic Tests at approximately 50%, Specialist Consultation at 47.44%, and General Consultation at 46.64%.

Reminder status remains associated with outcomes, but Week 6 adds an important validation: the cross-track Data Science analysis excluded 263 Cancelled appointments before comparing Attended versus No-show. Under that treatment, the reminder-related difference was approximately five percentage points rather than the stronger effect initially suggested. The conclusion is therefore refined: reminder status is a useful risk/segmentation variable, but the observed data do not establish causality.

2. WEEK 6 ANALYTICAL QUESTIONS

- Does previous no-show history remain associated with current no-shows?
- Does the reminder pattern remain meaningful when Cancelled appointments are treated separately?
- Which appointment types show the highest no-show rates?
- Are the Week 5 KPIs consistent with the deeper segmented analysis?
- Which findings have the greatest operational value for HealthConnect?

3. VALIDATED KPIS

KPI	Week 6 result	Interpretation
Attendance Rate	46.28%	Share of all appointments marked Attended.
No-Show Rate	48.46%	Share of all appointments marked No-show.
Follow-up No-Show Rate	51.23%	No-show rate for Follow-up appointments.
No-Show Rate — Attended vs No-Show	51.15%	No-show share after excluding Cancelled appointments from the denominator.

The 51.15% validation KPI is calculated only across Attended and No-show outcomes. It does not replace the 48.46% overall Week 5 baseline; it provides a directly comparable view for the cross-track analysis that excluded Cancelled appointments.

4. ADVANCED ANALYSIS FINDINGS

4.1 Previous No-Show History

The Power BI line chart shows an upward pattern in current no-show rate as the number of previous no-shows increases. The displayed series is approximately 43% at zero previous no-shows, 54% at one, 60% at two, 66% at three, 65% at four, and approximately 97% at five. The small highest-history group should be interpreted cautiously because extreme groups can contain fewer observations.

Decision implication: Previous no-show history is a strong candidate for targeted intervention and for consideration as a modelling feature in Data Science work.

4.2 APPOINTMENT TYPE

Appointment Type	No-Show Rate
Follow-up	51.23%
Diagnostic Test	≈50% (visual)
Specialist Consultation	47.44%
General Consultation	46.64%

Follow-ups are the highest-risk appointment type in the displayed dashboard, at 51.23%, which is 2.77 percentage points above the overall no-show rate of 48.46%. This makes follow-ups a practical segment for further operational investigation.

4.3 REMINDER CHANNEL

Appointment Type	Email	No Reminder	SMS	WhatsApp	Total
Diagnostic Test	46.27%	55.56%	47.11%	50.00%	49.75%
Follow-up	58.99%	53.13%	47.01%	53.35%	51.23%
General Consultation	45.37%	49.16%	45.21%	46.61%	46.64%
Specialist Consultation	42.00%	51.63%	43.97%	50.97%	47.44%
Total	48.41%	51.39%	45.75%	49.77%	48.46%

The segmented matrix shows that No Reminder has the highest overall no-show rate among the displayed reminder categories at 51.39%, compared with 48.41% for Email, 45.75% for SMS, and 49.77% for WhatsApp. However, channel performance varies by appointment type; for example, Email is 58.99% for Follow-ups while SMS is 47.01% for the same type. This variation argues against declaring a single channel universally superior.

4.4 CROSS-TRACK VALIDATION: CANCELLED APPOINTMENTS

The Data Science collaborator : Fatimah Odumyiwa identified a processing difference: 263 Cancelled appointments were excluded before their Attended-versus-No-show analysis, while the Data Analytics workflow retained them in the primary dataset. This difference changes the denominator and can affect the apparent no-show rate and reminder effect. The Week 6 dashboard therefore retains the full dataset for continuity while adding the 51.15% Attended-versus-No-show validation KPI.

This is a methodological validation rather than a correction of one track by the other. The evidence supports reporting both definitions clearly.

5. EVIDENCE-BASED BUSINESS INSIGHTS

- Repeated no-shows are a priority segment. Current no-show rates rise with previous no-show history, with an especially high displayed rate for patients with five previous no-shows.
- Follow-ups need targeted attention. Their 51.23% no-show rate is above the overall 48.46% rate.
- Reminder coverage matters, but channel choice is not uniform. No Reminder has the highest overall no-show rate in the dashboard, while the best-performing channel varies by appointment type.
- Outcome definition affects measured effect size. Excluding Cancelled appointments produces a different no-show/reminder estimate, so KPI definitions must be explicit.
- Distance and waiting time remain secondary. Week 5 analysis did not show meaningful relationships for these variables; Week 6 therefore does not elevate them to primary drivers.

6. BUSINESS RECOMMENDATIONS

- Prioritize repeat no-show patients: create a targeted intervention group based on previous no-show history.
- Strengthen reminder coverage: identify appointments with no recorded reminder and monitor attendance/no-show outcomes.
- Segment reminder strategy: evaluate reminder channels by appointment type instead of assuming one channel is best for every group.
- Investigate follow-ups: review scheduling, communication, patient context, and operational reasons for the elevated no-show rate.
- Standardize KPI definitions: report the full-outcome no-show rate separately from the Attended-versus-No-show rate.

7. LIMITATIONS

- The analysis is observational; associations do not prove that reminders or appointment type cause no-shows.
- Reminder-channel groups may differ in patient or appointment characteristics, creating confounding.
- The group with five previous no-shows may be relatively small, so its extreme rate should be validated on more observations.
- Missing-value treatment can introduce uncertainty, particularly for operational variables.
- The Week 6 analysis is decision support, not a production predictive model.

8. WEEK 7 ANALYTICAL TESTING REQUIREMENTS

- Validate the previous-no-show relationship on an additional or validation sample.
- Recalculate reminder performance using a consistent outcome definition and compare results with the cross-track analysis.
- Test whether the follow-up pattern persists across relevant patient and reminder segments.

- Monitor the four core KPIs over time and verify dashboard calculations against the source data.
- Provide validated candidate features and findings to Data Science and revise recommendations if new evidence contradicts them.

9. CONCLUSION

Week 6 strengthens the HealthConnect analysis by moving from broad Week 5 exploration toward validation and decision support. The most actionable evidence concerns previous no-show history, follow-up appointments, and reminder coverage. The cross-track review also demonstrates that analytical definitions matter: the treatment of Cancelled appointments materially changes the denominator and the estimated reminder relationship. HealthConnect should therefore use targeted segmentation and clearly defined KPIs while validating these findings before operational or predictive deployment.

APPENDIX — POWER BI ADVANCED DECISION SUPPORT SNAPSHOT

